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Effect of compound preparation *Tongqiao Jiannao* capsules on neural cell apoptosis and Bcl-2 and Bax protein levels in a rat model of brain ischemia/reperfusion injury★

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Abstract

BACKGROUND: Pharmacological studies have demonstrated that compound preparation *Tongqiao Jiannao* capsules composed of *Zexie, Baizhu, Honghua, Danshen*, and *Shexiang* can supplement *qi*, activate blood circulation, relieve blood stasis, induce resuscitation for alleviating pain, relieve pain, and dilate blood vessels.
OBJECTIVE: To observe the effects of *Tongqiao Jiannao* capsules on the levels of the anti-apoptotic protein Bcl-2 and the pro-apoptotic protein Bax, and verify the mechanism of action.
DESIGN, TIME AND SETTING: Randomized, controlled animal experiment, performed in the Laboratory of Biochemistry and Molecular Biology, Shanxi Medical University between June 2001 and December 2002.
MATERIALS: The right middle cerebral arteries of 24 healthy adult Sprague Dawley rats were occluded by the suture method. The primary Chinese herbal medicinal ingredients of *Tongqiao Jiannao* capsules are *Zexie, Baizhu, Honghua, Danshen*, and *Shexiang*, which were purchased from Shanxi Provincial Medicinal Material Company, China, and prepared into condensed granules in the Room for Chinese Herbal Medicine Preparation, Second Hospital, Shanxi Medical University. Bcl-2 and Bax immunohistochemical staining kits, a 3,3-diaminobenzidine(DAB) kit, and an *in situ* apoptosis detection kit were purchased from Wuhan Boster Bioengineering Co., Ltd., China.
METHODS: Twenty-four rats were randomly and evenly divided into three groups: (1) sham-operated rats in which sutures were inserted and immediately pulled out; (2) *Tongqiao Jiannao* capsule-treated rats that were intragastrically administered 6.5 g/kg/d *Tongqiao Jiannao* capsule preparation for seven successive days prior to middle cerebral artery occlusion (MCAO); and (3) MCAO rats without any other treatments.
MAIN OUTCOME MEASURES: The levels of neural cell apoptosis and Bcl-2 and Bax proteins at 24 hours post-surgery.
RESULTS: In the MCAO group, the numbers of apoptotic cells and Bax-positive cells were significantly increased, while the numbers of Bcl-2-positive cells were slightly decreased compared with the sham-operated group. Bcl-2- and Bax-positive cells and apoptotic cells were primarily distributed in the ischemic penumbra. In the *Tongqiao Jiannao* capsule-treated group, neuronal apoptosis was inhibited, and the number of Bcl-2-positive cells was significantly increased ($P < 0.01$), while the number of Bax-positive cells was significantly decreased ($P < 0.01$), compared with the MCAO group.
CONCLUSION: *Tongqiao Jiannao* capsules elevated Bcl-2 expression, lowered Bax expression, and inhibited cellular apoptosis during the process of cerebral ischemia/reperfusion injury.
Key Words: *Tongqiao Jiannao* capsules; cerebral ischemia/reperfusion; cellular apoptosis; Bcl-2; Bax; Bcl-2/Bax

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INTRODUCTION

Tongqiao Jiannao capsule preparation, a Chinese herbal compound preparation, is composed of *Zexie, Baizhu, Honghua, Danshen,* and *Shexiang*, and protects against cerebral ischemia. In the present study, a rat model of focal cerebral ischemia/reperfusion injury was used to observe the effects of *Tongqiao Jiannao* capsules on neural cell apoptosis and Bcl-2 and Bax protein levels after cerebral ischemia, to investigate the mechanism of action[1-2].

MATERIALS AND METHODS

Materials

This study was performed in the Laboratory of Biochemistry and Molecular Biology, Shanxi Medical University between June 2001 and December 2002. Twenty-four healthy, male, Sprague Dawley rats of clean grade, aged 2–3 months, weighing 220–250 g, were provided by the Laboratory Animal Center of Shanxi Medical University (Permission No. 010101). The following protocol complied with guidelines issued by the Shanxi Provincial Ethics Committee for the care and use of laboratory animals.

Drugs and reagents: Chinese herbal medicinal materials of *Tongqiao Jiannao* capsules were purchased from Shanxi Provincial Medicinal Material Company, China and prepared into condensed granules in the Room for Chinese Herbal Medicine Preparation, Second Hospital, Shanxi Medical University. Before use, *Tongqiao Jiannao* capsules were prepared to the required concentration with physiological saline. Bcl-2 and Bax immunohistochemical staining kits, 3,3-diaminobenzidine(DAB) kit, and terminal deoxynucleotidyl transferase -mediated dUTP nick end labeling (TUNEL) kit were purchased from Wuhan Boster Bioengineering Co., Ltd., China.

Methods

Grouping

Twenty-four rats were randomly and evenly divided into three groups: sham-operated, *Tongqiao Jiannao* capsule-treated and middle cerebral artery occlusion (MCAO) groups.

Preparation of models

After anesthesia by an intraperitoneal injection of 10% chloral hydrate (4 mg/kg), rat MCAO was performed using the suture method [3]. One hour later, sutures were taken out to allow blood flow recovery. In the sham-operated group, sutures were inserted and immediately taken out. The *Tongqiao Jiannao* capsule-treated group was intragastrically administered 6.5 g/kg/d Chinese herbal medicine for seven successive days. In addition, one hour after intragastric administration on the day MCAO was performed. During the surgery, rectal temperature was kept constant at 36.5–37.5 ℃. Subsequent to recovering consciousness, functional neurological deficits were scored. A score of 0 indicated no neurological deficits; a score of 1 (failure to fully extend opposite forepaw) correlated with mild focal neurological deficits; a score of 2 (rotating to the left) correlated with moderate focal neurological deficits; and a score of 3 (falling to the opposite side) correlated with severe focal deficits. Rats with a score of 4 did not walk spontaneously and had a reduced level of consciousness. Rats with a score of more than 1 point were included in the following experiment.

Preparation of specimens

After ischemia for 1 hour and reperfusion for 24 hours, successfully modeled rats were transcardially perfused with 200 mL of physiological saline. Subsequent to anesthesia, fresh brain tissue was taken out, postfixed with 4% paraformaldehyde, routinely dehydrated with ethanol, cleared, embedded with paraffin, serially sliced at 5-μm thickness, and finally, TUNEL and immunohistochemically stained.

Detection of indices

Apoptotic neural cells were detected by TUNEL assay. Levels of Bcl-2 and Bax protein were determined by immunohistochemistry. Known, unstained positive control sections were used. Under 400-fold magnification, five visual fields were randomly selected from the peri-ischemic area to count positive cells.

Statistical analysis

All data were statistically analyzed by Rui Wang and Guanglai Li using SPSS 11.5 software, and are expressed as Mean ± SD. One-way analysis of variance and least significant difference tests were used. A probability value of less than 0.01 was considered statistically significant.

RESULTS

Quantitative analysis of experimental animals

A total of 24 rats were included in the final analysis.

Detection of cellular apoptosis by TUNEL assay

Apoptotic cells exhibited brown nuclei, but were stained unevenly, and chromatin was decentralized towards the periphery. A small number of TUNEL-positive neural cells were found in the sham-operated group. In the MCAO and *Tongqiao Jiannao* capsule-treated groups, TUNEL-positive cells were primarily distributed in the peri-ischemic penumbra rather than in the ischemic core area. A comparison of the numbers of TUNEL-positive cells is shown in Table 1.

Table 1 Comparisons of TUNEL-, Bcl-2- and Bax-positive cells among different groups ($\bar{x}\pm s$, $n = 8$)

Group	TUNEL	Bcl-2	Bax	Bcl-2/Ba
Sham-operated	10.38±3.38	3.00±1.07	5.25±1.67	0.61±0.23
MCAO	34.50±3.21^a	20.38±2.45^a	37.63±2.13^a	0.54±0.07
Tongqiao Jiannao capsule treated	23.75±3.20^b	38.88±3.00^b	25.13±2.59^b	1.56±0.19^b

TUNEL: terminal deoxynucleotidyl transferase-mediated dUTP nick end labeling; MCAO: middle cerebral artery occlusion; $^aP < 0.01$, *vs.* sham-operated group; $^bP < 0.01$, *vs.* MCAO group

Detection of Bcl-2 and Bax expression by immunohistochemistry

Bcl-2- and Bax-immunopositivity manifested as brown-stained cytoplasm. Bcl-2- and Bax-positive cells were primarily distributed in the peri-ischemic area, in accordance with the distribution of apoptotic cells in the same area. A comparison of the numbers of Bcl-2- and Bax-positive cells is shown in Table 1.

DISCUSSION

Pharmacological studies have demonstrated that the simple Chinese herbal recipe *Zhexie* can be used to promote diuresis, lower blood pressure and blood glucose levels; *Baizhu* can boost immunity, provide resistance to aging, cancer, oxidation, and blood coagulation, decrease blood lipid and glucose levels, inhibit bacteriosis, and promote diuresis in addition to affecting digestive and cardiovascular systems; *Honghua* (Flos Carthami) can be used to promote blood circulation, remove blood stasis and restore normal menstruation; *Danshen* can apparently assist recovery from ischemia/reperfusion injury to the heart, liver, lung, and brain; *Shexiang* can induce resuscitation, activate blood circulation, and relieve blood stasis. Therefore, compound preparation of *Tongqiao Jiannao* capsules composed of the above-mentioned simple recipes can supplement *qi*, activate blood circulation, relieve blood stasis, induce resuscitation for alleviating pain, relieve pain, and dilate blood vessels.

Studies have demonstrated that cerebral ischemia/reperfusion injury is closely related to cellular apoptosis[4-5]. Neuronal apoptosis is one of important mechanisms of delayed neural cell injury subsequent to cerebral ischemia. In addition, neuronal apoptosis is influenced by several factors, such as ions, apoptosis-associated protein Bcl-2/Bax, and Fas/Fasl[6-9]. Increasing attention has been paid to reducing neuronal apoptosis by many pathways and exerting neuroprotective effects[10-18]. It has been shown that Bcl-2, as an important endogenous anti-apoptotic factor, plays a critical role in the regulation of neuronal apoptosis, and that it is of significance for the maintenance of neural cell survival following cerebral ischemia/reperfusion. Increases in the levels of Bcl-2 gene expression and protein can inhibit cellular apoptosis and boost cell survival [19-22]. Bax is another important member of the Bcl-2 family, and is highly homologous to Bcl-2 gene. Bax and Bcl-2 have opposite effects. Bcl-2/Bax heterodimers are necessary for Bcl-2 to exert its anti-apoptotic effects. Over-expression of Bax can antagonize the inhibitory effects of Bcl-2 on cellular apoptosis. Therefore, the ratio between Bcl-2 and Bax determines cellular survival or apoptosis subsequent to cerebral ischemia/reperfusion injury. Neuronal apoptosis can be inhibited by regulating Bcl-2 and Bax gene expression, resulting in neuroprotective effects [23-30].

The present experimental results demonstrated that, compared with the sham-operated group, the apoptosis rate was obviously increased in the MCAO group. In addition, positive cells were primarily found in the peri-infarct ischemic penumbra. These results further confirm that cell apoptosis is indeed involved in cerebral ischemia/reperfusion injury. Simultaneously, Bax levels were increased. Bcl-2- and Bax-positive cells were distributed similarly to apoptotic cells. These results demonstrate that when cerebral ischemia/reperfusion injury occurs, Bax expression is increased and Bax homodimer promotes cellular apoptosis. In addition, up-regulation of Bcl-2 expression inhibits cellular apoptosis, reflecting a physiologically protective mechanism. However, the ratio between Bcl-2 and Bax levels is decreased due to the increase in Bax expression, leading to an apparent increase in the number of apoptotic cells in the MCAO group. Compared with the MCAO group, neuronal apoptosis was inhibited, the number of Bcl-2-positive cells showed and obvious increase, and the number of Bax-positive cells was decreased in the *Tongqiao Jiannao* capsule-treated group. These results indicate that the neuroprotective effects of *Tongqiao Jiannao* capsules on cerebral ischemia/reperfusion injury are associated with an anti-apoptotic effect and an increase in the number of Bcl-2-positive cells, a decrease in the number of Bax-positive cells, and an enhancement of the ratio between Bcl-2 and Bax levels.

The present study only investigated the effects of compound preparation *Tongqiao Jiannao* capsules administered intragastrically from the standpoint of cellular apoptosis and Bcl-2 and Bax levels, and the effective ingredients of the Chinese herbal material were not analyzed. Thus, the effects of individual components can not be precisely elucidated.

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